Linux System Resource Monitoring Tool

Develop a Bash script to monitor CPU utilization, memory usage, running processes, and disk usage. Generate a system resource usage report.

Google Cloud Shell-style terminal

ILLUSTRATIVE VIEW

Cloud Shell Editor | monitor.sh | lines 1-46

Cloud Shell-style demonstration

SOURCE FILE

BASH SOURCE

```bash
#!/usr/bin/env bash
# Linux System Resource Monitoring Tool
# Collects CPU, memory, process, and disk statistics and writes a text report.

set -Eeuo pipefail
export LC_ALL=C

REPORT_DIR="${REPORT_DIR:-./reports}"
SAMPLE_INTERVAL="${SAMPLE_INTERVAL:-1}"

usage() {
  cat <<USAGE
Linux System Resource Monitoring Tool

Usage: $0 [--help]

Environment variables:
  REPORT_DIR       Directory where reports are saved (default: ./reports)
  SAMPLE_INTERVAL  Seconds between CPU samples (default: 1)

Example:
  REPORT_DIR=./reports SAMPLE_INTERVAL=1 ./monitor.sh
USAGE
}

if [[ "${1:-}" == "--help" || "${1:-}" == "-h" ]]; then
  usage
  exit 0
fi

if [[ $# -gt 0 ]]; then
  echo "Error: unexpected argument: $1" >&2
  usage >&2
  exit 2
fi

for cmd in awk date df head hostname mkdir ps sleep tee uname; do
  if ! command -v "$cmd" >/dev/null 2>&1; then
    echo "Error: required command '$cmd' is not installed." >&2
    exit 1
  fi
done

if [[ ! "$SAMPLE_INTERVAL" =~ ^[0-9]+([.][0-9]+)?$ ]] || ! awk -v n="$SAMPLE_INTERVAL" 'BEGIN { exit !(n > 0) }'; then
  echo "Error: SAMPLE_INTERVAL must be a positive number of seconds." >&2
  exit 2
```

Cloud Shell-style layout; the script and displayed run are from a Linux execution.

Google Cloud Shell-style terminal

ILLUSTRATIVE VIEW

Cloud Shell Editor | monitor.sh | lines 47-92

Cloud Shell-style demonstration

SOURCE FILE

BASH SOURCE

```
047  fi
048
049  cpu_snapshot() {
050    local _ user nice system idle iowait irq softirq steal _rest
051    read -r _ user nice system idle iowait irq softirq steal _rest < /proc/stat
052    local total=$((user + nice + system + idle + iowait + irq + softirq + steal))
053    local idle_total=$((idle + iowait))
054    printf '%s %s\n' "$total" "$idle_total"
055  }
056
057  read -r total_before idle_before < <(cpu_snapshot)
058  sleep "$SAMPLE_INTERVAL"
059  read -r total_after idle_after < <(cpu_snapshot)
060
061  delta_total=$((total_after - total_before))
062  delta_idle=$((idle_after - idle_before))
063  if (( delta_total > 0 )); then
064    cpu_usage=$(awk -v total="$delta_total" -v idle="$delta_idle" 'BEGIN { printf "%.1f%%", (total-idle)*100/total }')
065  else
066    cpu_usage="N/A"
067  fi
068
069  read -r mem_total_kb mem_available_kb < <(
070    awk '/^MemTotal:/ { total=$2 } /^MemAvailable:/ { available=$2 } END { print total+0, available+0 }' /proc/meminfo
071  )
072  if (( mem_total_kb > 0 )); then
073    mem_used_kb=$((mem_total_kb - mem_available_kb))
074    mem_usage_pct=$(awk -v used="$mem_used_kb" -v total="$mem_total_kb" 'BEGIN { printf "%.1f%%", used*100/total }')
075    mem_total_gib=$(awk -v v="$mem_total_kb" 'BEGIN { printf "%.2f GiB", v/1048576 }')
076    mem_used_gib=$(awk -v v="$mem_used_kb" 'BEGIN { printf "%.2f GiB", v/1048576 }')
077    mem_available_gib=$(awk -v v="$mem_available_kb" 'BEGIN { printf "%.2f GiB", v/1048576 }')
078  else
079    mem_usage_pct="N/A"
080    mem_total_gib="N/A"
081    mem_used_gib="N/A"
082    mem_available_gib="N/A"
083  fi
084
085  process_table=$(ps -eo pid,comm,%cpu,%mem --sort=-%cpu | head -n 6)
086  disk_line=$(df -hP / | awk 'NR==2 { printf "%s  %s total  %s used  %s available  %s used  mounted on %s", $1, $2, $3, $4, $5, $6 }')
087
088  hostname_value=$(hostname 2>/dev/null || printf 'unknown')
089  kernel_value=$(uname -r 2>/dev/null || printf 'unknown')
090  os_value="Linux"
091  if [[ -r /etc/os-release ]]; then
092    os_value=$(awk -F= '/^PRETTY_NAME=/ { gsub(/^"|"$/, "", $2); print $2; exit }' /etc/os-release)
```

Cloud Shell-style layout; the script and displayed run are from a Linux execution.

Google Cloud Shell-style terminal

ILLUSTRATIVE VIEW

Cloud Shell Editor | monitor.sh | lines 93-136

Cloud Shell-style demonstration

SOURCE FILE

BASH SOURCE

```
093     os_value="${os_value:-Linux}"
094   fi
095   cpu_model=$(awk -F: '/^model name[[:space:]]*:/ { sub(/^[ \t]+/, "", $2); print $2; exit }' /proc/cpuinfo 2>/dev/null || true)
096   cpu_model="${cpu_model:-Not reported by system}"
097
098   mkdir -p "$REPORT_DIR"
099   timestamp=$(date '+%Y%m%d_%H%M%S')
100   report_file="${REPORT_DIR%/}/system_resource_report_${timestamp}.txt"
101
102   {
103     echo "========================================================================"
104     echo "             LINUX SYSTEM RESOURCE MONITORING REPORT"
105     echo "========================================================================"
106     printf 'Generated on : %s\n' "$(date '+%A, %d %B %Y at %H:%M:%S %Z')"
107     printf 'Hostname     : %s\n' "$hostname_value"
108     printf 'Operating OS : %s\n' "$os_value"
109     printf 'Kernel       : %s\n' "$kernel_value"
110     echo
111     echo "1. CPU UTILIZATION"
112     echo "------------------------------------------------------------------------"
113     printf 'CPU model             : %s\n' "$cpu_model"
114     printf 'Average CPU utilization: %s (sample interval: %s second(s))\n' "$cpu_usage" "$SAMPLE_INTERVAL"
115     echo
116     echo "2. MEMORY UTILIZATION"
117     echo "------------------------------------------------------------------------"
118     printf 'Total memory          : %s\n' "$mem_total_gib"
119     printf 'Memory in use         : %s\n' "$mem_used_gib"
120     printf 'Memory available      : %s\n' "$mem_available_gib"
121     printf 'Memory utilization    : %s\n' "$mem_usage_pct"
122     echo
123     echo "3. TOP RUNNING PROCESSES (sorted by CPU usage)"
124     echo "------------------------------------------------------------------------"
125     printf '%s\n' "$process_table"
126     echo
127     echo "4. ROOT FILESYSTEM DISK USAGE"
128     echo "------------------------------------------------------------------------"
129     printf '%s\n' "$disk_line"
130     echo
131     echo "========================================================================"
132     echo "Report generated successfully."
133     echo "========================================================================"
134   } | tee "$report_file"
135
136   printf '\nReport saved to: %s\n' "$report_file"
```

Cloud Shell-style layout; the script and displayed run are from a Linux execution.

Google Cloud Shell-style terminal

ILLUSTRATIVE VIEW

Cloud Shell terminal | ./monitor.sh

Cloud Shell-style demonstration

TERMINAL

RUN OUTPUT

```
$ SAMPLE_INTERVAL=1 REPORT_DIR=./reports ./monitor.sh

================================================================
            LINUX SYSTEM RESOURCE MONITORING REPORT
================================================================
Generated on : Monday, 02 November 2026 at 07:20:48 UTC
Hostname     : 43289f3b8982
Operating OS : Debian GNU/Linux 13 (trixie)
Kernel       : 6.18.57

1. CPU UTILIZATION
----------------------------------------------------------------
CPU model             : AMD EPYC 9V74 80-Core Processor
Average CPU utilization: 0.4% (sample interval: 1 second(s))

2. MEMORY UTILIZATION
----------------------------------------------------------------
Total memory          : 5.81 GiB
Memory in use         : 1.12 GiB
Memory available      : 4.69 GiB
Memory utilization    : 19.3%

3. TOP RUNNING PROCESSES (sorted by CPU usage)
----------------------------------------------------------------
    PID COMMAND         %CPU %MEM
    336 artifact_tool_r  6.0  4.3
    319 python           2.3  2.3
    300 python           1.2  1.2
    296 python           0.6  0.6
      1 supervisord      0.3  0.4

4. ROOT FILESYSTEM DISK USAGE
----------------------------------------------------------------
overlay  32G total  14M used  30G available  1% used  mounted on /

================================================================
Report generated successfully.
================================================================

Report saved to: ./reports/system_resource_report_20261102_072048.txt
```

Cloud Shell-style layout; the script and displayed run are from a Linux execution.

Google Cloud Shell-style terminal

ILLUSTRATIVE VIEW

Cloud Shell terminal | saved report

Cloud Shell-style demonstration

TERMINAL

REPORT FILE

```
$ cat reports/system_resource_report_20261102_072048.txt

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